

THE MASTER COMPENDIUM

Electrical Machines II - Induction Machine Mastery

Phase 6

The Starting Problem and Practical Solutions

This phase explains why an induction motor draws a huge current at starting and how engineers reduce that current without destroying the starting torque requirement.

Focus: DOL starting, stator-resistance starting, autotransformer starting, star-delta starting, slip-ring rotor resistance starting, SQIM vs SRIM, deep-bar rotor, and double-cage rotor.

Prepared for semester examination revision.

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1 Where Phase 6 Fits in the Roadmap

The exact mental position of Phase 6

Phase 4 gave the equivalent circuit and Phase 5 converted that circuit into power and torque. Phase 6 asks a very practical question:

What happens at the instant of starting, when $s = 1$?

At starting, the rotor is stationary. Hence the slip is maximum, rotor frequency is maximum, and the motor draws a very high current. The purpose of a starter is not to start the motor magically; it is to start the motor **safely** by controlling current, voltage, and torque.

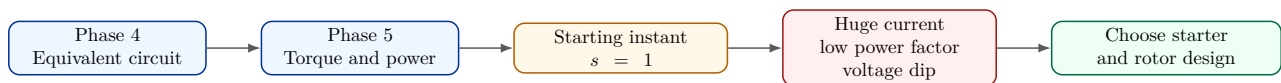


Figure 1: Roadmap continuity: Phase 6 begins by putting $s = 1$ in the circuit and torque equations.

What Phase 6 must teach you

After this phase, you should be able to write exam answers on:

1. why starting current is high in an induction motor;
2. why starting torque falls as square of applied voltage;
3. DOL, stator-resistance, autotransformer and star-delta starting;
4. slip-ring motor starting by external rotor resistance;
5. difference between squirrel-cage and slip-ring induction motors;
6. deep-bar and double-cage rotor action using skin effect and leakage reactance.

2 The Starting Problem from the Equivalent Circuit

2.1 The physical reason: no speed, no back action

In a running induction motor, the rotor rotates close to synchronous speed. Therefore the relative speed between the rotating magnetic field and rotor conductors is small. The induced rotor frequency is also small:

$$f_2 = sf_1.$$

At starting,

$$N_r = 0, \quad s = \frac{N_s - N_r}{N_s} = 1, \quad f_2 = f_1.$$

So the rotor behaves like the short-circuited secondary of a transformer. Since the rotor branch impedance is small, the stator draws a very large current from the supply.

Simple language answer

At starting, the rotor is not moving, so the stator field cuts the rotor conductors at maximum relative speed. Maximum relative speed produces maximum rotor emf. The rotor circuit is short-circuited, so the rotor current becomes high. To balance the rotor ampere-turns, the stator draws a high current from the supply.

2.2 Correct standstill circuit

At starting, $s = 1$. Therefore the mechanical load resistance vanishes:

$$R'_2 \left(\frac{1-s}{s} \right) = R'_2 \left(\frac{1-1}{1} \right) = 0.$$

So the rotor branch is only $R'_2 + jX'_2$, referred to the stator side. The circuit must be joined with a proper bottom return path.

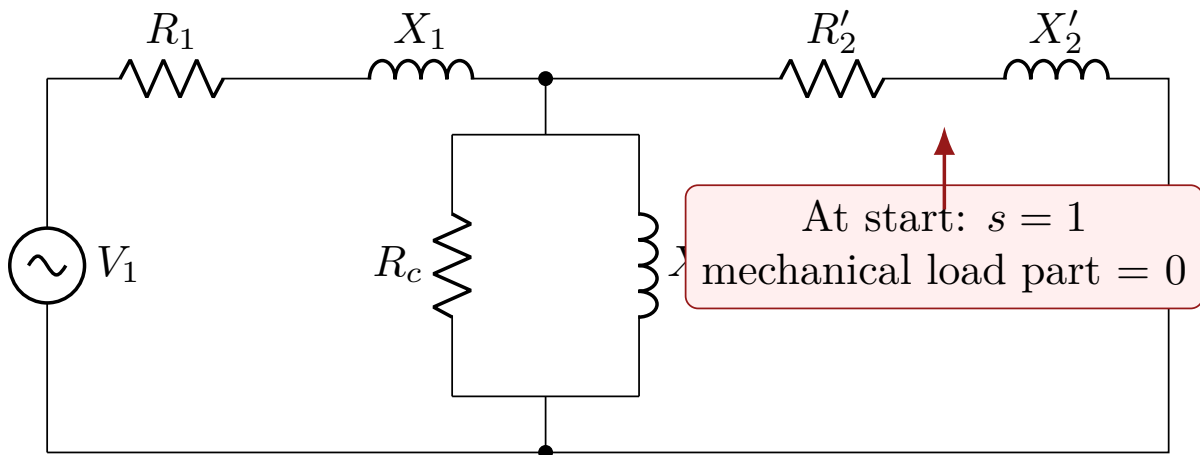


Figure 2: Standstill equivalent circuit. The rotor branch is closed; the mechanical conversion resistance is zero at $s = 1$.

2.3 Approximate starting current and starting torque

Ignoring the shunt branch for a simple exam derivation:

$$Z_{st} = (R_1 + R'_2) + j(X_1 + X'_2).$$

$$I_{st} = \frac{V_1}{\sqrt{(R_1 + R'_2)^2 + (X_1 + X'_2)^2}}$$

The starting torque is obtained from the Phase 5 torque equation by putting $s = 1$:

$$T_{st} = \frac{3}{\omega_s} \frac{V_1^2 R'_2}{(R_1 + R'_2)^2 + (X_1 + X'_2)^2}$$

Starting torque is proportional to V_1^2

If the applied stator voltage is reduced to kV_1 , then approximately:

$$I_{st,new} = kI_{st}, \quad T_{st,new} = k^2 T_{st}.$$

This one square-law relation is the heart of voltage-reduction starters.

3 Direct-On-Line Starting

3.1 Concept

In direct-on-line starting, the motor terminals are directly connected to the rated three-phase supply. There is no voltage reduction and no external impedance during starting.

DOL in one line

DOL gives the simplest and strongest starting action, but it also draws the highest starting current. It is acceptable only when the supply system and the motor rating can tolerate the inrush current.

3.2 Correct DOL power circuit

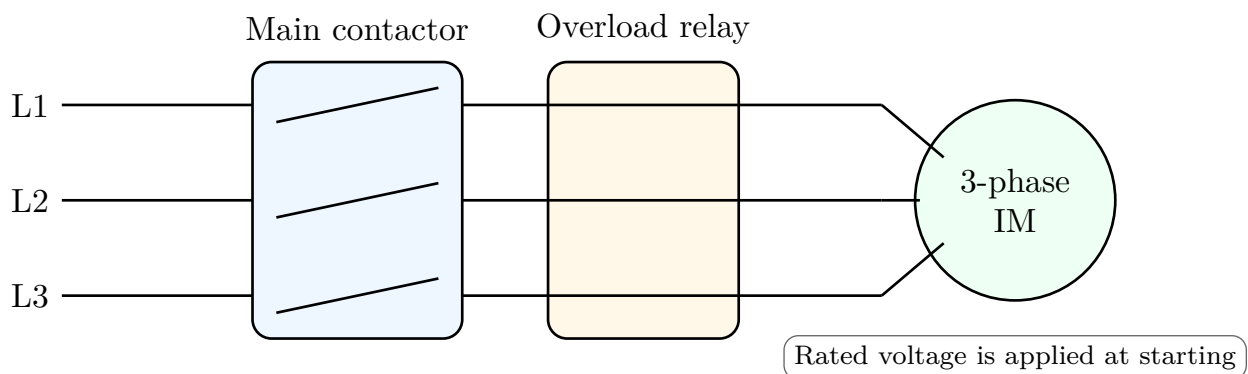


Figure 3: DOL starter power circuit. The lines are continuous from supply, through contactor and overload relay, to the motor.

3.3 Mathematical result

For DOL starting:

$$V_{motor} = V_{rated}, \quad I_{st} = I_{DOL}, \quad T_{st} = T_{DOL}.$$

Typical starting current may be several times the rated current because the standstill impedance is small.

Advantages and disadvantages of DOL starter

Advantages	Simple, cheap, reliable, full starting torque, less maintenance.
Disadvantages	Very high starting current, supply voltage dip, mechanical jerk, not suitable for large motors on weak supply systems.

4 Stator-Resistance Starting

4.1 Concept

A resistance is inserted in series with each stator phase during starting. This drops part of the supply voltage, so the motor receives a reduced voltage. Once the motor accelerates, the resistance is gradually cut out.

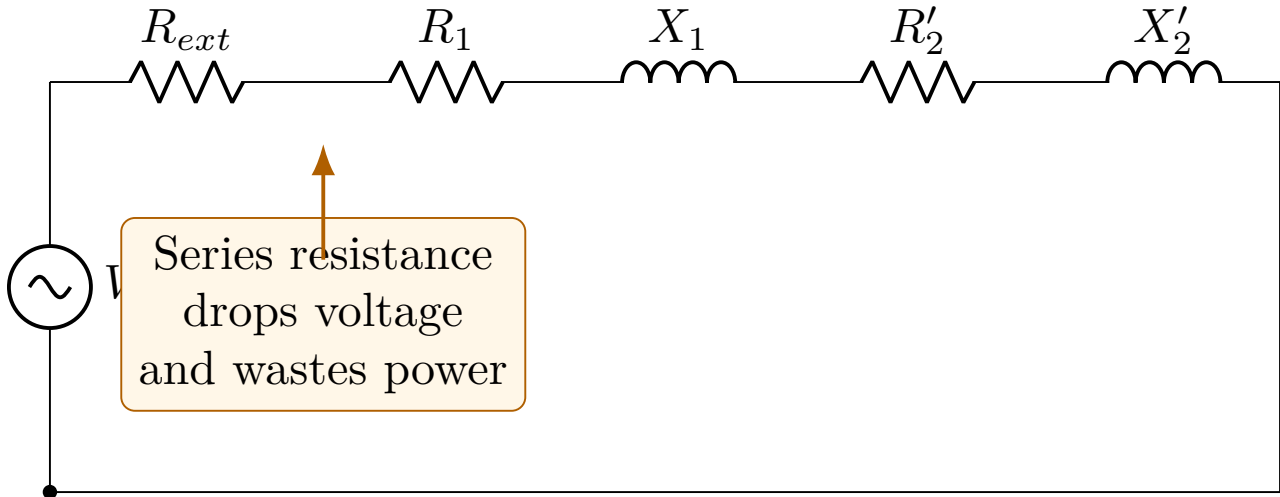


Figure 4: Approximate per-phase stator-resistance starting circuit. The return path is closed; the added resistance is in series with stator and rotor impedances.

4.2 Effect on current and torque

Let the motor terminal voltage become kV_1 , where $0 < k < 1$. Then:

$$\boxed{I_{st} \approx kI_{DOL}}, \quad \boxed{T_{st} \approx k^2T_{DOL}}.$$

Why this method is not very efficient

The external stator resistance reduces starting current, but it also wastes energy as heat. More importantly, the torque falls as the square of voltage. So a large reduction in current causes a very large reduction in starting torque.

Exam sentence

Stator resistance starting is simple but inefficient because it reduces the applied motor voltage and wastes power in the external resistors. Since $T \propto V^2$, the starting torque becomes poor when current is reduced considerably.

5 Autotransformer Starting

5.1 Concept

An autotransformer starter applies a reduced voltage to the motor during starting. After the motor accelerates, the autotransformer is cut out and the motor is connected directly to the supply.

Let

$$k = \frac{V_{motor}}{V_{line}}$$

be the autotransformer tap ratio during starting.

5.2 Correct joined diagram

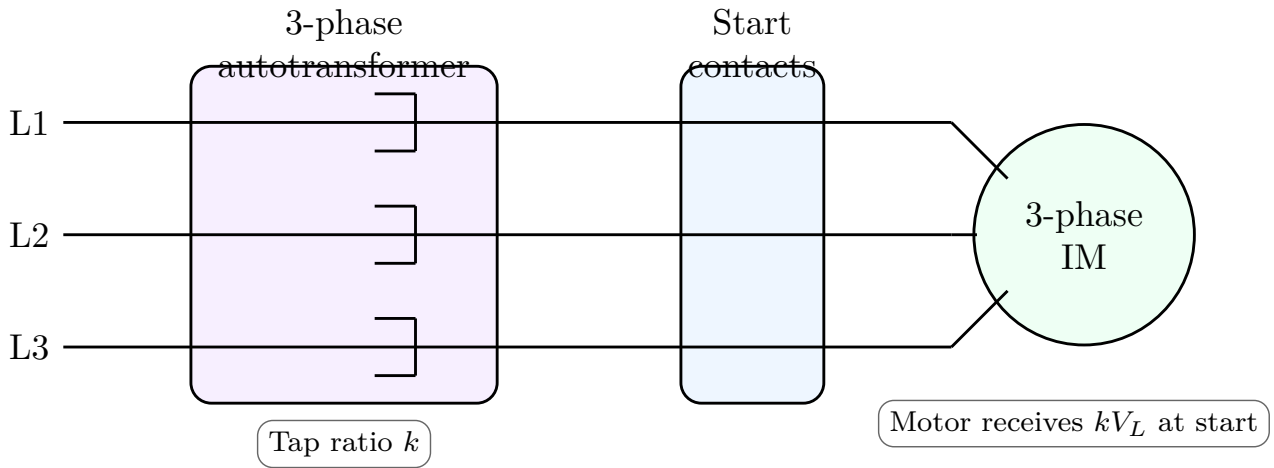


Figure 5: Autotransformer starter. The motor receives reduced voltage through a tapped autotransformer during starting.

5.3 Derivation of current and torque ratios

Motor voltage during starting:

$$V_{motor} = kV_{line}.$$

So motor current becomes approximately:

$$I_{motor,st} = kI_{DOL}.$$

Due to transformer action, the supply current is further reduced:

$$I_{line,st} = kI_{motor,st} = k^2I_{DOL}.$$

Torque depends on motor voltage squared:

$$T_{st} = k^2T_{DOL}.$$

$$\boxed{\frac{T_{st}}{T_{DOL}} = k^2}, \quad \boxed{\frac{I_{line,st}}{I_{DOL}} = k^2}, \quad \boxed{\frac{I_{motor,st}}{I_{DOL}} = k}.$$

For the same line current drawn from the supply, the autotransformer starter gives better starting torque than simple stator-resistance starting.

6 Star-Delta Starting

6.1 Concept

Star-delta starting is used for motors that are designed to run normally in delta. During starting, the stator winding is connected in star so each phase receives a smaller voltage. After the motor accelerates, the winding is reconnected in delta for normal running.

Important condition

Star-delta starting is suitable only when the motor is designed to run in delta at the given line voltage. It cannot be used blindly for every induction motor.

6.2 Star and delta connection diagrams

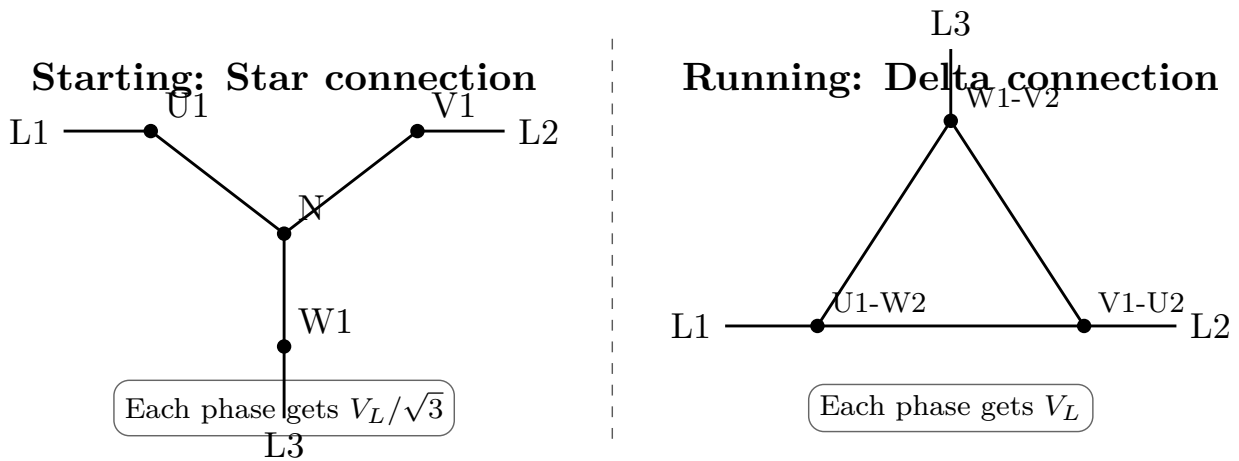


Figure 6: Star-delta idea. Start in star to reduce phase voltage; run in delta to apply rated phase voltage.

6.3 Derivation: why current and torque become one-third

For the same line voltage V_L :

$$\text{In delta running: } V_{ph,\Delta} = V_L.$$

$$\text{In star starting: } V_{ph,Y} = \frac{V_L}{\sqrt{3}}.$$

Since torque is proportional to square of phase voltage,

$$\frac{T_Y}{T_\Delta} = \left(\frac{V_{ph,Y}}{V_{ph,\Delta}} \right)^2 = \left(\frac{1}{\sqrt{3}} \right)^2 = \boxed{\frac{1}{3}}.$$

For current, let Z_{st} be standstill phase impedance.

$$I_{ph,\Delta} = \frac{V_L}{Z_{st}}, \quad I_{L,\Delta} = \sqrt{3}I_{ph,\Delta} = \sqrt{3}\frac{V_L}{Z_{st}}.$$

In star:

$$I_{L,Y} = I_{ph,Y} = \frac{V_L/\sqrt{3}}{Z_{st}}.$$

Therefore:

$$\boxed{\frac{I_{L,Y}}{I_{L,\Delta}} = \frac{1}{3}}, \quad \boxed{\frac{T_Y}{T_{\Delta}} = \frac{1}{3}}.$$

Most important star-delta result

$$\boxed{I_{line,start} = \frac{1}{3} I_{line,DOL \text{ in delta}}}, \quad \boxed{T_{start} = \frac{1}{3} T_{DOL \text{ in delta}}}.$$

Hence star-delta starting is not suitable for loads requiring high starting torque.

7 Comparison of Voltage-Reduction Starters

Starter	Motor voltage at start	Line current from supply	Starting torque
DOL	V	I_{DOL}	T_{DOL}
Stator resistance	kV approximately	nearly kI_{DOL}	k^2T_{DOL}
Autotransformer	kV	k^2I_{DOL}	k^2T_{DOL}
Star-delta	$V/\sqrt{3}$ per phase	$I_{DOL}/3$ compared with delta DOL	$T_{DOL}/3$

Table 1: Comparison of common reduced-voltage starters. Here $0 < k < 1$.

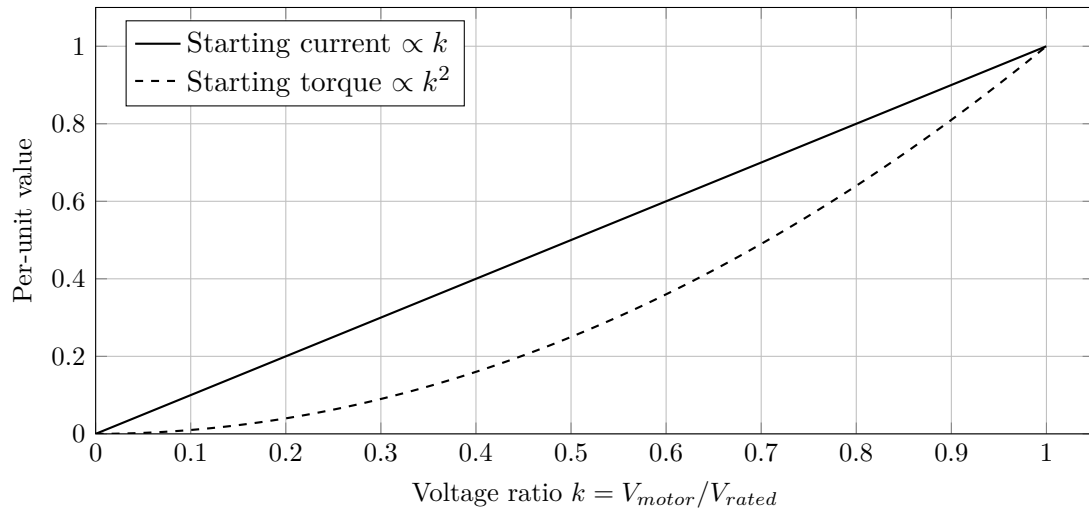


Figure 7: Voltage reduction reduces current linearly but reduces torque quadratically. This graph explains why heavy loads cannot be started with too low a voltage.

Practical selection logic

Do not select a starter only by current reduction. Always check starting torque. A motor may draw a safe current but fail to accelerate if the starting torque is below load torque.

8 Slip-Ring Induction Motor Starting by External Rotor Resistance

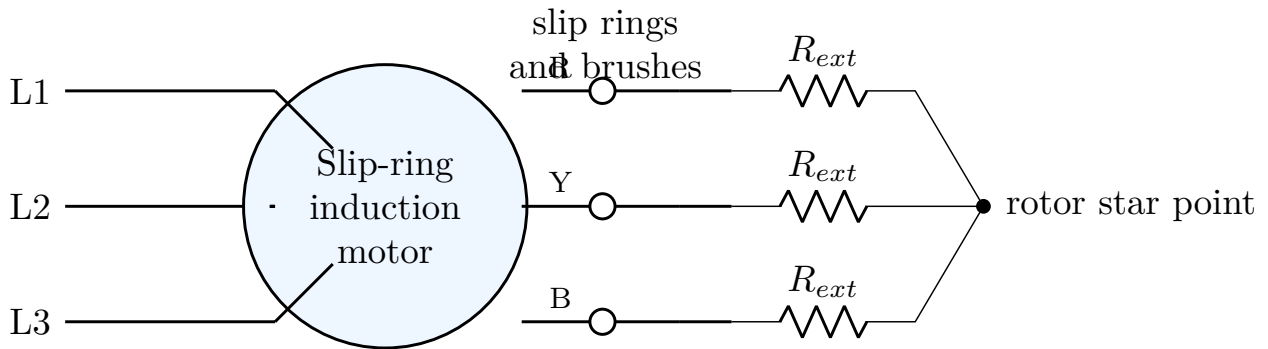
8.1 Why this method is special

In a squirrel-cage motor, the rotor bars are permanently short-circuited. We cannot insert external resistance in the rotor circuit. But in a slip-ring induction motor, rotor winding terminals are brought out through slip rings and brushes. External resistances can be inserted during starting.

Core advantage

External rotor resistance increases starting torque and also limits starting current. This is why slip-ring induction motors are useful for high starting-torque loads such as cranes, hoists and elevators.

8.2 Correct rotor-resistance starter diagram



External rotor resistance is cut out step-by-step as speed rises

Figure 8: Slip-ring induction motor starting. The rotor circuit is accessible, so external resistances can be inserted safely during starting.

8.3 Torque effect of adding rotor resistance

At starting, $s = 1$. If external resistance referred to the stator side is R'_{ext} , then the effective rotor resistance becomes:

$$R'_{2,total} = R'_2 + R'_{ext}.$$

The starting torque becomes approximately:

$$T_{st} = \frac{3}{\omega_s} \frac{V_1^2 R'_{2,total}}{(R_1 + R'_{2,total})^2 + (X_1 + X'_2)^2}.$$

Increasing rotor resistance initially increases starting torque. Maximum starting torque occurs when the effective rotor resistance approximately satisfies:

$$R'_{2,total} \approx X_1 + X'_2 \quad \text{if } R_1 \text{ is neglected.}$$

More exactly:

$$R'_{2,total} = \sqrt{R_1^2 + (X_1 + X'_2)^2}.$$

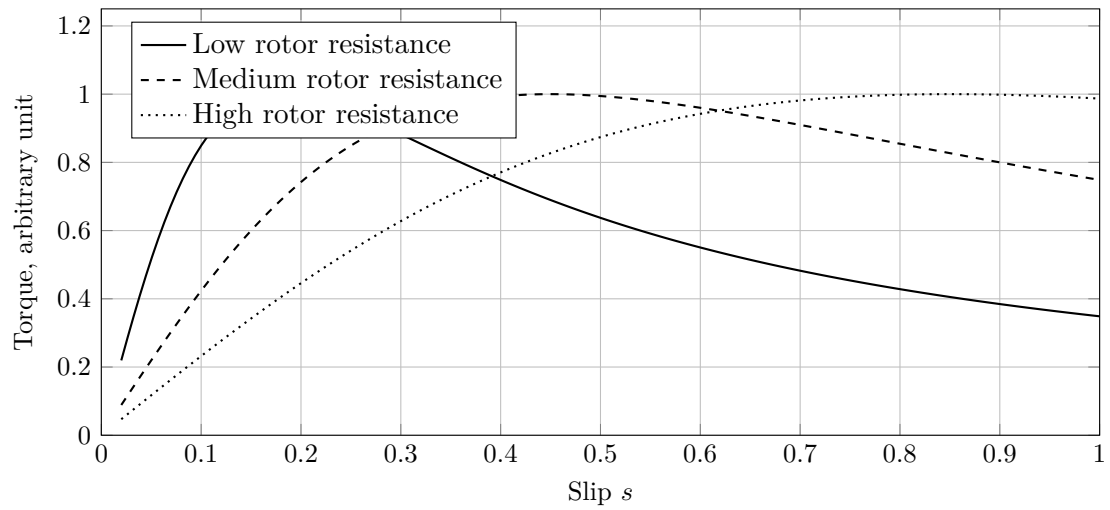


Figure 9: Increasing rotor resistance shifts the maximum-torque point toward higher slip. For starting, this means the high-torque point can be moved close to $s = 1$.

9 Squirrel-Cage Motor vs Slip-Ring Motor

Point	Squirrel-cage induction motor (SQIM)	Slip-ring induction motor (SRIM)
Rotor construction	Conducting bars short-circuited by end rings	Three-phase wound rotor connected to slip rings
External rotor resistance	Not possible	Possible using slip rings and brushes
Starting current	High	Can be controlled
Starting torque	Moderate unless special rotor design is used	High starting torque possible
Maintenance	Low, rugged, cheap	Higher due to brushes and slip rings
Efficiency and reliability	Generally high and reliable	Slightly lower reliability due to moving contacts
Applications	Pumps, fans, blowers, general drives	Cranes, hoists, elevators, heavy starting loads

Table 2: SQIM vs SRIM from starting point of view.

Exam-style conclusion

A squirrel-cage motor is simple, robust and cheap but gives limited control over starting current and starting torque. A slip-ring motor is costlier and needs more maintenance, but it allows external rotor resistance, giving high starting torque with controlled starting current.

10 Deep-Bar Rotor

10.1 Why deep bars are used

A deep-bar rotor is a squirrel-cage rotor whose bars are made deep inside the slot. It uses skin effect to automatically change rotor resistance with speed.

Skin effect in simple words

When rotor frequency is high, current does not distribute uniformly through the whole conductor. It crowds near the upper part of the bar. This reduces the effective conducting area and increases effective resistance.

At starting:

$$s = 1, \quad f_2 = f_1 \quad (\text{high rotor frequency}).$$

So current crowds near the top of the deep bar. Effective resistance is high, giving better starting torque.

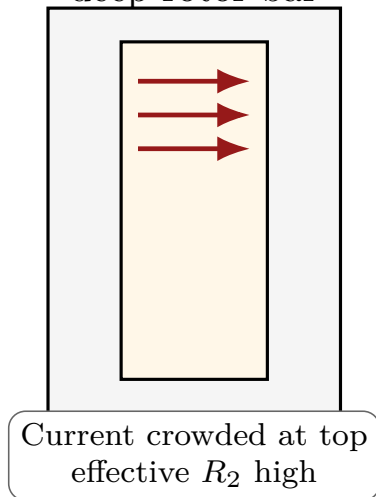
At normal running:

$$s \ll 1, \quad f_2 = sf_1 \quad (\text{low rotor frequency}).$$

Current uses almost the full depth of the bar. Effective resistance is low, giving good efficiency.

10.2 Deep-bar current distribution diagram

Starting condition: f_2 high
deep rotor bar



Running condition: f_2 low
same bar at low frequency

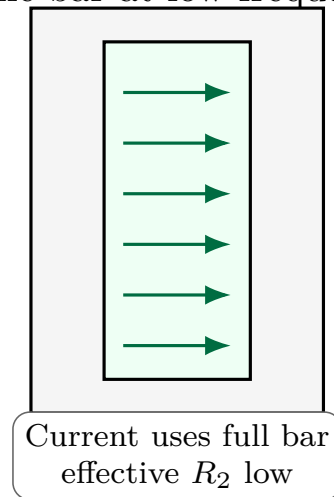


Figure 10: Deep-bar rotor action. The same conductor behaves like high resistance at starting and low resistance at running.

bar. The effective area decreases and rotor resistance increases. Increased rotor resistance improves starting torque. At normal speed, rotor frequency is low, current penetrates the full bar, resistance becomes low, and efficiency improves.

11 Double-Cage Rotor

11.1 Construction and purpose

A double-cage rotor has two cages in each rotor slot:

1. **Outer cage:** high resistance and low leakage reactance.
2. **Inner cage:** low resistance and high leakage reactance.

At starting, rotor frequency is high. The inner cage has high reactance, so most current flows in the outer cage. Because the outer cage has high resistance, starting torque improves.

At normal running, rotor frequency is low. The reactance of the inner cage becomes small, so most current flows through the low-resistance inner cage. Efficiency becomes good.

11.2 Double-cage cross-section

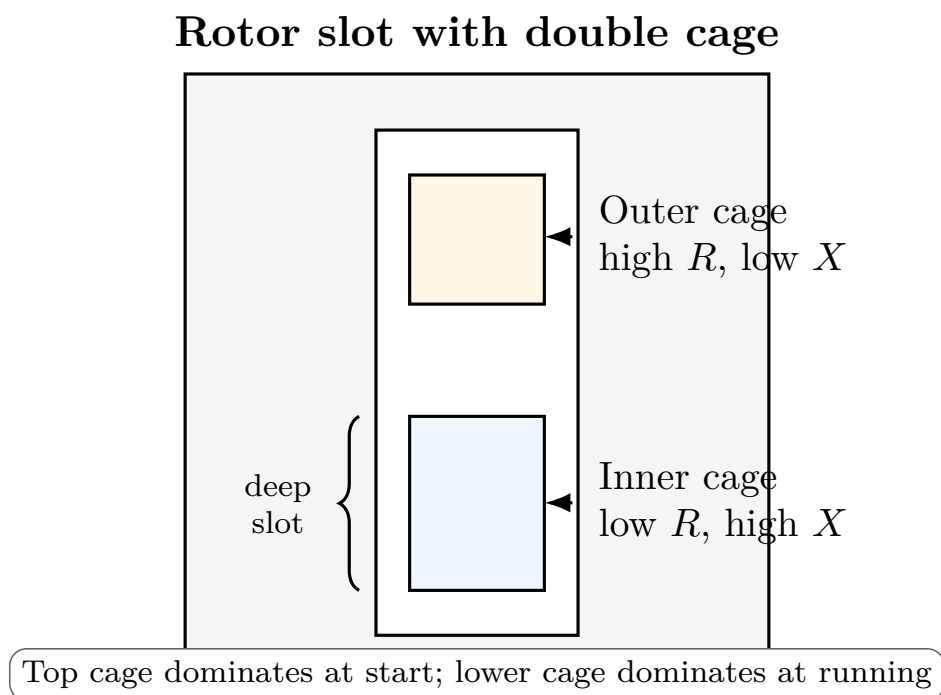


Figure 11: Physical idea of a double-cage rotor. Two cages give two different electrical behaviours.

11.3 Correct parallel equivalent circuit

The outer and inner cages are electrically in parallel because both cages experience induced emf and both are connected through end rings. A clean equivalent rotor circuit is therefore:

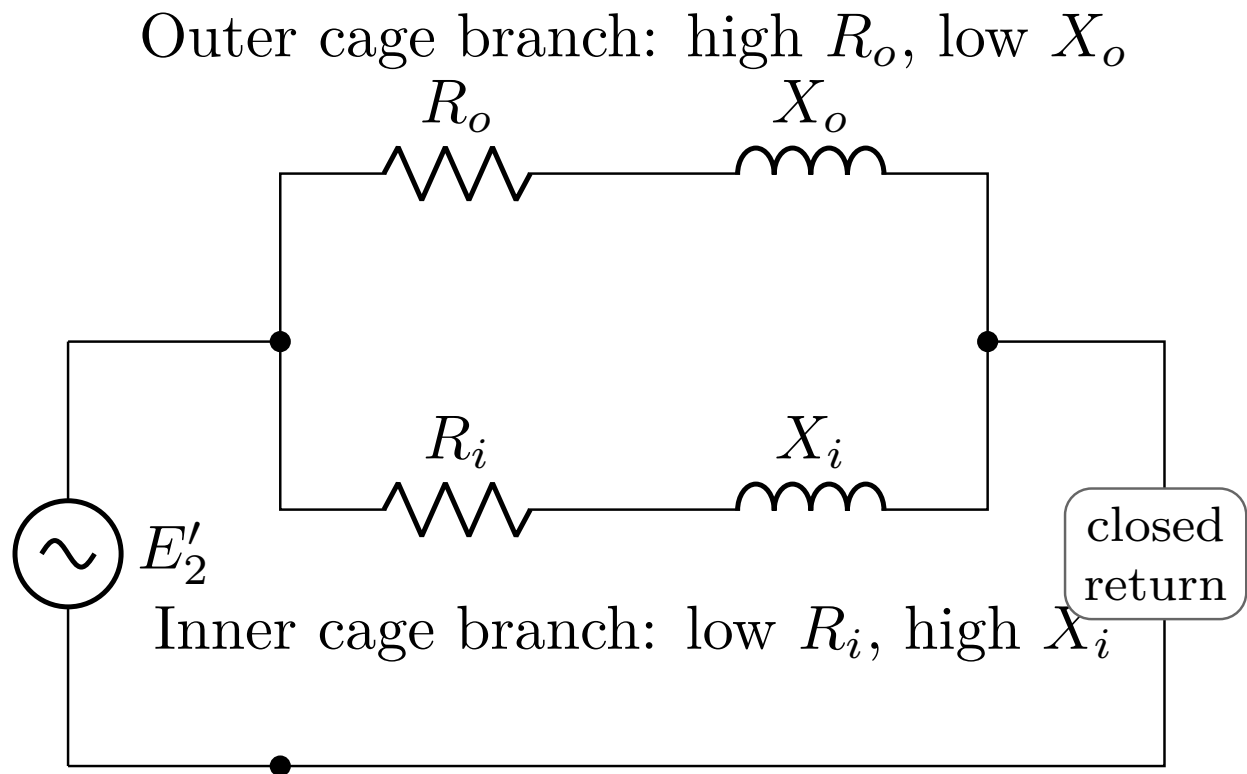


Figure 12: Correct double-cage rotor equivalent circuit. The two cage branches are parallel and are joined by a common return path.

$$Z_o = R_o + jX_o, \quad Z_i = R_i + jX_i,$$

$$Z_{rotor} = \frac{Z_o Z_i}{Z_o + Z_i}.$$

At starting, X_i is high, so inner-cage current is restricted. At running, rotor frequency is low, so X_i falls and the low-resistance inner cage carries more current.

12 Torque-Speed Behaviour of Different Starting Arrangements

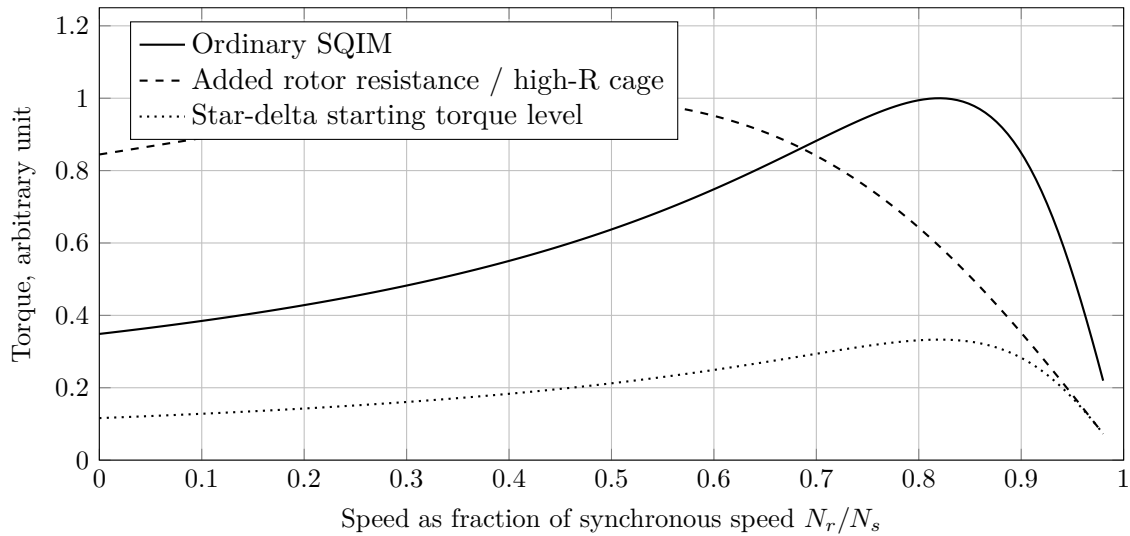


Figure 13: Starting methods change either voltage level or effective rotor resistance. Voltage reduction lowers the curve; rotor-resistance control shifts the high-torque region toward starting.

How to read this graph

At zero speed, $N_r/N_s = 0$ and $s = 1$. A starter must provide enough torque at the left side of the graph to overcome load torque and accelerate the motor. If starting torque is below load torque, the motor will not pick up speed.

13 Worked Numerical Pattern for Star-Delta and Autotransformer Starting

Problem

A delta-connected induction motor has DOL starting current $6I_{FL}$ and DOL starting torque $1.8T_{FL}$. Find the starting current and starting torque when:

1. the motor is started by star-delta starter;
2. the motor is started by an autotransformer starter with tap ratio $k = 0.65$.

Solution

Given:

$$I_{DOL} = 6I_{FL}, \quad T_{DOL} = 1.8T_{FL}.$$

1. Star-delta starting

$$I_Y = \frac{1}{3}I_{DOL} = \frac{1}{3}(6I_{FL}) = 2I_{FL}.$$

$$T_Y = \frac{1}{3}T_{DOL} = \frac{1}{3}(1.8T_{FL}) = 0.6T_{FL}.$$

So star-delta gives a safe current but weak starting torque.

2. Autotransformer starting, $k = 0.65$

$$I_{motor} = kI_{DOL} = 0.65(6I_{FL}) = 3.9I_{FL}.$$

$$I_{line} = k^2I_{DOL} = 0.65^2(6I_{FL}) = 2.535I_{FL}.$$

$$T_{st} = k^2T_{DOL} = 0.65^2(1.8T_{FL}) = 0.7605T_{FL}.$$

Final answer:

$$I_Y = 2I_{FL}, \quad T_Y = 0.6T_{FL}$$

$$I_{line,AT} = 2.535I_{FL}, \quad I_{motor,AT} = 3.9I_{FL}, \quad T_{AT} = 0.7605T_{FL}.$$

Important interpretation

For almost the same supply-line current, the autotransformer starter gives more starting torque than star-delta. This is why autotransformer starting is preferred when a better starting torque per supply current is required.

14 Exam-Oriented Answer Templates

14.1 Question: Why does an induction motor draw high starting current?

Answer template

At starting, the rotor speed is zero. Hence slip is:

$$s = \frac{N_s - 0}{N_s} = 1.$$

Rotor frequency becomes:

$$f_2 = sf_1 = f_1.$$

The relative speed between stator RMF and rotor conductors is maximum, so rotor induced emf is maximum. Since the rotor circuit is nearly short-circuited, rotor current becomes high. The stator must supply this rotor current by transformer action, so the stator input current becomes very large. This causes voltage dip, heating and mechanical shock. Therefore starters are used to limit starting current.

14.2 Question: Compare DOL, star-delta and autotransformer starters.

Answer template

Method	Principle	Current	Torque
DOL	Direct rated voltage to motor	highest	full DOL torque
Star-delta	start in star, run in delta	$I_{DOL}/3$	$T_{DOL}/3$
Autotransformer	reduced voltage using tap k	$k^2 I_{DOL}$ from supply	$k^2 T_{DOL}$

14.3 Question: Explain double-cage rotor.

Answer template

A double-cage rotor has two cages: an outer cage of high resistance and low leakage reactance, and an inner cage of low resistance and high leakage reactance. At starting, rotor frequency is high, so the inner cage reactance is high and current mainly flows in the high-resistance outer cage. This gives good starting torque. At normal running, rotor frequency is low, so the inner cage reactance falls and current flows mainly through the low-resistance inner cage. This gives good efficiency during normal operation. Thus a double-cage rotor combines high starting torque with good running efficiency.

15 One-Page Memory Map

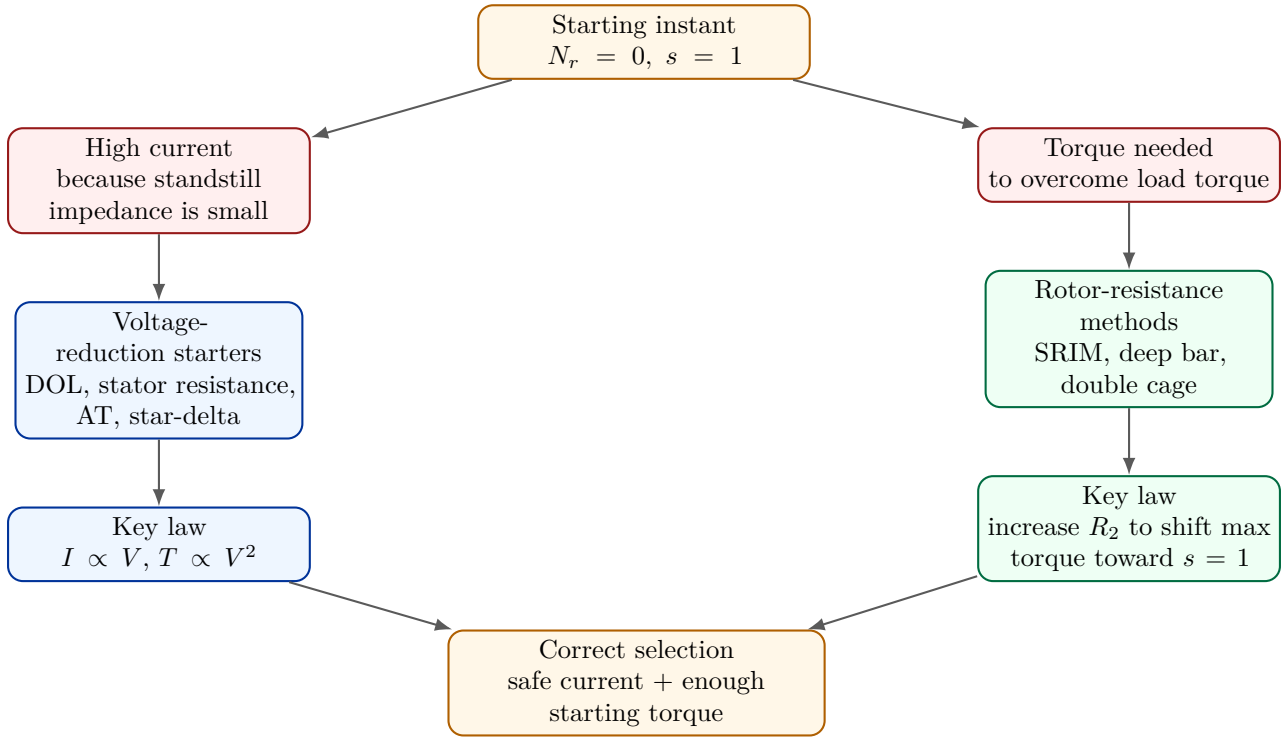


Figure 14: Phase 6 memory map. Never separate current control from torque requirement.

$$s = 1 \quad \text{at starting,} \quad f_2 = f_1,$$

$$I_{st} = \frac{V_1}{\sqrt{(R_1 + R'_2)^2 + (X_1 + X'_2)^2}},$$

$$T_{st} = \frac{3}{\omega_s} \frac{V_1^2 R'_2}{(R_1 + R'_2)^2 + (X_1 + X'_2)^2},$$

$$V \rightarrow kV \quad \Rightarrow \quad I \rightarrow kI, \quad T \rightarrow k^2 T,$$

$$\text{Star-delta: } I_{line} = I_{DOL}/3, \quad T = T_{DOL}/3,$$

$$\text{Autotransformer: } I_{line} = k^2 I_{DOL}, \quad T = k^2 T_{DOL}.$$

End of Phase 6